Pale Cyst Nematode (*Globodera pallida*): A Comprehensive Review of Biology, Impact, Management, and Data Resources

I. Executive Summary

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The pale cyst nematode (PCN) *Globodera pallida* (Stone, 1973) Behrens, 1975 is an invasive plant nematode that threatens potato production worldwide. Native to South America, this microscopic roundworm has spread to multiple agricultural regions through the movement of contaminated soil and infected plant material. The nematode feeds on plant roots and disrupts plant growth, leading to the reduction of crop production. The most concerning characteristic of this pest is its ability to survive in soil for extremely long periods inside protective cysts. Agricultural authorities report that the PCN can remain in soil for decades because the cyst protects the eggs from environmental stress; however, this pest can remain dormant and viable in soil for 30 years.[4] Potatoes are a major global food crop, infestations of the pale cyst nematode represent a serious threat to agricultural production and food security. This review summarizes current knowledge regarding biology, identification, host range, global distribution, and management strategies associated with the PCN.

II. Introduction

The PCN is a soil-dwelling roundworm that infects plant roots and interferes with plant growth. This species is considered one of the most damaging pests that affect potato production.

The nematode was first discovered in the United States in Idaho in 2006.[4] This discovery led to strict quarantine and monitoring programs designed to prevent the pest from spreading to other regions that produce potatoes. The PCN is described as a soil dwelling roundworm that feeds on the roots of high value crops such as potatoes, tomatoes, and eggplants.[4] Damage occurs when the nematode feeds on plant roots and disrupts the plant's ability to absorb water and nutrients from the soil. In heavily infested areas, crop yields may decline significantly causing major economic losses because if PCN populations are left unchecked, they can reduce crop yields by 20 to 70 percent.[4] Since they can survive in soil for long periods of time, eradication programs require long term monitoring and strict regulatory programs.

III. Biology and Identification of the Pale Cyst Nematode (*Globodera pallida*)

Life Cycle: Egg, Juvenile (Nymph Equivalent), and Adult Stages

The PCN goes through three stages during its life cycle, which includes, egg, juvenile, and adult. Eggs are produced by female nematodes and remain protected in a hardened cyst within the soil. When conditions are favorable in terms of weather and host plant nematodes are released at a juvenile state. Juveniles then begin their search on the roots of the host plant and feed to develop into adults.

Physical Description and Distinguishing Features

- **Adults:** Adult female nematodes appear as small white or pale-yellow spheres attached to plant roots. As they mature and die, the body darkens and forms a brown cyst containing eggs. Adult males remain as a worm and leave the root after maturity.
- **Eggs:** Eggs remain extremely small and remain inside the cyst, and each cyst contains hundreds of eggs which can remain viable for many years in soil.
- **Juvenile:** Juveniles are microscopic and maintain a worm like appearance. They move through the soil pores and use chemical cues from plant roots to locate host plants.[1]

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Host Plants and Feeding Behavior

The PCN is limited to infecting plants within the Solanaceae family. The most important host is the potato, but other plants such as tomatoes and eggplants may also be affected.[8] The PCN feeds by inserting its stylet into plant cells and withdrawing nutrients during this, the nematode alters plant cells to form specialized feeding structures. Plant pathology describes this as the nematode induces specialized feeding cells within the host root that provide nutrients for its development.[2] This process then weakens the root system and reduces the plant's ability to absorb water and nutrients.

Geographical Distribution (Global and United States)

The PCN is native to South America but has worldwide distribution in potato-producing regions.[8] It is also known to occur in Newfoundland, Canada and Idaho. In Canada, related cyst nematode species such as the golden nematode is known to occur in Newfoundland, Vancouver Island, Quebec and Alberta.[7] Invasive species reports indicate that the pest spreads primarily through soil movement associated with agriculture, the primary pathway for spread is the movement of contaminated soil attached to equipment or plant material.[5]

Table 1: Key Characteristics of Pale Cyst Nematode Life Stages

Life Stage	Size (approx.)	Color/Appearance	Key Distinguishing Features	Typical Location
Eggs	Microscopic	Transparent	Protected inside cyst	Soil
Juvenile (J2)	~0.5mm	Worm-like transparent body	Infective stage that searches for roots	Soil near roots
Adult Females	~0.4-0.5mm	White to pale-yellow, later brown	Body swells and becomes cyst	Attached to root of plant
Adult Males	~1mm	Slender worm-like body	Leaves root after maturity to fertilize females	Soil around roots
Cysts	Small spherical body	Brown hardened shell	Dead female body containing eggs	Soil and root surfaces

IV. Disease

Causative Agents

The PCN damages plants primarily through feeding and altering root cell structures to form a feeding site. These structures divert nutrients from the plant to the nematode and weaken plant growth. Limited information is available regarding interactions with other microorganisms.

Mechanism of Transmission by PCN

The PCN spreads primarily through the movement of contaminated soil. The soil contains the cysts that may attach to other equipment and/or plant materials. Programs emphasize the importance of sanitation practices to prevent the spread of PCN. This includes making sure all farm machinery, trucks, construction, and other equipment are clean when going from field to field. Do not allow equipment in your fields unless it has been thoroughly cleaned.[4] The pest may also spread through infected plant material, contaminated seeds, and soil.

Symptoms PCNs leave on Plants

- **Leaf Symptoms:** Heavy infestations may cause yellowing leaves, wilting, and poor plant growth across affected areas of the field. In some cases, patches of plants may appear stunted or weakened compared to surrounding crops.[8]
- **Differentiation from Nutritional Deficiencies:** Nematode damage usually appears in irregular patches across the field rather than evenly across the entire crop.
- **Root Symptoms:** Infected plants may develop poorly formed root systems due to the nematode feeding on them. Small white females and cysts may sometimes become visible on the surface of infected roots during the late stage of plant development.

Disease Progression and Economic Impact

Progression begins with small patches of weakened plants in infested fields. As the PCN population increases, damage becomes more severe and crop growth declines. Economic loss associated with the pale cyst nematode can be substantial with uncontrolled infestations reducing up to 80% of crops.[5] This can result in trade restrictions and quarantine measures that affect agricultural markets.

Global Distribution of Disease (Global and United States)

As strict quarantine methods have been placed to keep the pale cyst nematode combined. Limited information is available regarding global distribution beyond major potato growing regions.

V. Management and Control Strategies

Integrated Pest Management

Integrated pest management strategies combined with several methods which include crop rotation, sanitation, and chemical control. These were designed to reduce nematode populations while minimizing environmental impact.

Monitoring and Early Detection Methods

Monitoring and early detection is important in preventing the spread of the PCN. Preventive measures include cleaning farm machinery, and equipment before moving between

fields to prevent contaminated soil from spreading. Farmers are also advised to use certified potato seeds that come fields that are tested and confirmed to be free of PCN.[4]

Chemical Control Approaches

Chemical control methods include treating PCN infested field with a chemical fumigant in the late summer or early fall and sometimes in early spring, depending on soil temperature.[4] However, chemical fumigants are toxic to nontarget organisms and cause other environmental harm.[1] Some of the most effective chemicals have been banned due to this and growers must adapt and use alternative methods such as rotation, biocontrol, and resistance.[1]

Biological Control

Biological control includes several approaches that use natural organisms or biological processes to reduce nematode populations. These control methods include natural enemies, trap crops, and induced hatching methods. Natural enemies such as fungi can attack nematodes directly by producing nematocidal compounds, colonizing plant roots, or parasitizing nematode eggs.[1] Another biological method used are trap crops to control the population. These plants stimulate nematode eggs to hatch but do not allow them to reproduce and are described as plants that stimulate hatch of *G. pallida* but are non-hosts that prevent reproduction.[3]

One studied trap crop is the *Solanum sisymbriifolium*, which has been shown to significantly reduce nematode populations. Experiments have reported that the use of this plant was able to reduce more than 97% of the nematode populations.[6] Another strategy involves stimulating the eggs to hatch when no host plant is present. Sometimes referred to as a "suicide hatch," causes juveniles to emerge from the cyst but prevents them from completing their life cycle.[1] Studies report that they can induce up to 80% hatching of *G. pallida* eggs in the absence of the host plant.[1]

Cultural Control Practices

Cultural control methods include crop rotation, certified seeds, cleaning farm equipment, and avoiding soil movement. The basic methods are used to reduce and stop the spread of the nematode population. Crop rotation with nonhost plants is one of the most effective practices because it reduces the ability of the nematode to reproduce.

Regulatory Frameworks and Quarantine Programs

Since 2006, the United States and Canada have been practicing a joint surveillance program to monitor potato cyst nematodes (pale cyst nematode) in all fields associated with seed potatoes traded between both countries.[8] PCNs are regulated by the United States Department of Agriculture (USDA) to prevent the spread from current infested areas such as Idaho and New York.[7]

VI. Available Datasets and Resources

Image Datasets for PCN and Disease Detection

- Several research institutions maintain image collections of pale cyst nematodes and infected plant root for educational purposes. Limited images are available currently for disease detection.

Genetic and Genomic Datasets

- Limited Information—Research on this topic for the pale cyst nematode is still ongoing or insufficient.

Population Monitoring and Disease Incidence Data

- Limited Information—Research in this topic for the pale cyst nematode is still ongoing or insufficient.

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VII. Ongoing Research and Future Prospects

Current research is focused on biological control methods that improve the host plants resistance and identify environmentally friendly alternatives to chemical nematicides. Scientists are also studying genetic resistance in potato crops as well as technology for early detection of infestations. Limited information is available regarding long-term global research efforts.

VIII. Conclusion

The PCN is a serious agricultural pest that threatens potato production. Its ability to survive for decades in soil and reproduce within protective cysts making it extremely difficult to control once established. Understanding the biology, distribution, and management of this pest is essential for protecting crop production. Continued monitoring and research will be necessary to prevent the spread of this invasive species and reduce economic impact.

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